

Phase 1 — SQL Analytics Layer (Business Intelligence Foundation)

Business Goal

Create a reliable, queryable hospital data layer that leadership can trust for operational and financial decision-making.

Technical Objectives

Transform raw CSV files into a structured relational database, enforce data integrity, and create reusable business views for analysis and reporting.

Tasks

- Create a new database schema for the hospital system.
- Define SQL tables: patients, visits, and billing with appropriate data types.
- Define primary keys and foreign key relationships between tables.
- Document which queries benefit from which indexes.
- **Operational Analysis**
 - Retrieve the **top 10 departments by total visit volume**.
 - Identify the **top 5 departments with the highest average length of stay**.
 - Find the **percentage of High Risk visits per department**.
 - Determine the **average number of visits per patient by city**.
 - Identify **doctors handling the highest number of High Risk visits**.
- **Financial Analysis**
 - Retrieve the **top 10 insurance providers by total billed amount**.
 - Identify the **top 5 insurance providers with the highest claim rejection rate**.
 - Find the **average payment delay (payment_days) by insurance provider**.
 - Calculate the **revenue realization ratio (approved_amount / billed_amount) by department**.
 - Identify **visits where billed_amount is high but approved_amount is zero or missing**.
- **Data Quality and Integrity Checks**
 - Detect **visits that do not have a corresponding billing record**.
 - Detect **billing records that do not have a corresponding visit**.
 - Identify **patients with duplicate patient_id values**.
 - Find records with **missing or invalid length_of_stay_hours or payment_days**.
 - Identify **visits linked to patients with missing insurance provider information**.